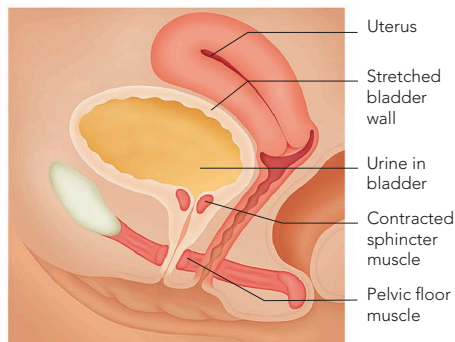


INCONTINENCE

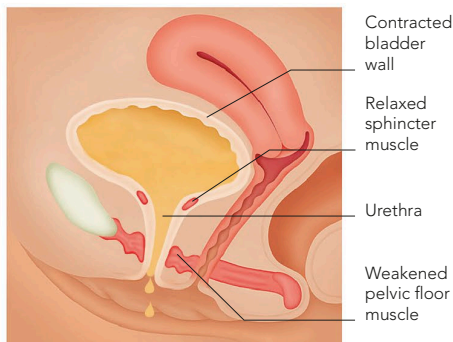
A tendency to leak urine, incontinence most commonly occurs in women, elderly people, and those with brain or spinal cord damage. Women after childbirth may be susceptible because their pelvic floor muscles may be weak. There are

different types, such as stress incontinence (see below). In urge incontinence, irritable bladder muscle causes the bladder to contract and expel all its urine. In total incontinence, a nervous system disorder such as multiple sclerosis causes total loss of bladder function.



NORMAL BLADDER

A healthy bladder expands like a balloon as it fills with urine. The sphincter muscles and surrounding pelvic floor muscles keep the exit closed. Nerve signals from stretch sensors in the bladder wall travel to the brain, signalling the need for emptying.



STRESS INCONTINENCE

To empty the bladder, the sphincter and pelvic floor muscles relax, and the detrusor muscle in the bladder wall contracts, forcing urine along the urethra. In incontinence, weak muscles may allow this to happen without proper control, so urine leaks out.

KIDNEY STONES

Kidney stones are solid, mineral-rich objects formed from chemicals, such as calcium salts, in urine. They can take years to form, and grow in various shapes and sizes. A stone may stay in the kidney and cause few problems, but it can increase the risk of urinary tract infection.

CRYSTALS

Kidney stones are usually formed from the mineral salt calcium oxalate, when it crystallizes from the urine. Crystals of this salt are shown here.



WHERE KIDNEY STONES FORM

Kidney stones can occur in any of the urine-collecting parts of the kidneys, such as the calyx or the renal pelvis.

